Experiment No. – 7						
Date of Performance:	**9/4/2026**					
Date of Submission:	**16/4/2026**					
Conceptual Understanding (Cognitive) (20%) (02)	Code Execution (Psychomotor) (15%) (1.5)	Problem-Solving & Debugging (Cognitive + Psychomotor) (20%) (02)	Professional Conduct & Lab Ethics (Affective) (20%) (02)	Output / Result Accuracy (Product) (15%) (1.5)	Documentation & Result Analysis (Cognitive) (10%) (01)	Sign with Date

Experiment No. 7

1.1 Aim: Design and Deploy a chain code and deploy in Hyperledger Fabric

1.2 Course Outcome: Develop of application using permissioned Hyperledger Fabric network.

1.3 Learning Objectives: Develop a chain code in Hyperledger Fabric.

1.4 Requirement: To perform the design and deployment of chaincode in Hyperledger Fabric, the following requirements are needed:

- Operating System: Linux environment (Ubuntu via WSL)
- Platform: Hyperledger Fabric framework
- Tools & Software: Git, Curl, Docker, Docker Compose, Go language
- Hardware Requirements: Minimum 4GB RAM, dual-core processor, stable internet connection
- Development Environment: Terminal (WSL) for executing Fabric commands
- Fabric Components: Fabric binaries, Docker images, and sample network (fabric-samples)

- Dependencies: jq (for JSON processing), proper environment variables setup

1.5 Related Theory: Hyperledger Fabric is an open-source blockchain framework developed under the Linux Foundation. It is designed for building permissioned blockchain networks, where all participants are known and verified, making it suitable for enterprise applications.

1. Blockchain in Hyperledger Fabric
Blockchain is a distributed ledger technology where data is stored in blocks that are linked together. In Hyperledger Fabric:
- Transactions are recorded securely and immutably
- All participants maintain a synchronized copy of the ledger
- It ensures transparency, security, and trust

Unlike public blockchains, Fabric is permissioned, meaning only authorized users can access the network.

2. Key Components of Hyperledger Fabric
- Peer Nodes:
  These nodes maintain the ledger and execute chaincode. Each organization in the network has its own peers.
- Orderer Node:
  Responsible for ordering transactions and creating blocks before adding them to the ledger.
- Channel:
  A private communication layer between organizations. Only members of a channel can access its data.
- Ledger:
  Stores all transactions and current state (world state) of assets.

3. Chaincode (Smart Contract)
Chaincode is the business logic of the blockchain network. It defines:
- How assets are created, updated, and queried
- Rules for transactions

In this experiment, a basic chaincode (asset-transfer-basic) is deployed using Go language, which manages asset data like ID, owner, and value.

4. Working of the Fabric Network
The working process includes:
1. Network Initialization:
   The network is started using Docker containers which create peers and orderer.
2. Channel Creation:
   A channel is created to enable communication between organizations.
3. Chaincode Deployment:
   Chaincode is installed and instantiated on the channel.

4. Transaction Execution:
   Users invoke chaincode functions to perform operations like adding or querying assets.
5. Ledger Update:
   Transactions are validated, ordered, and committed to the ledger.

5. Role of Docker in Fabric
Docker is used to:
- Run peer and orderer nodes as containers
- Maintain isolated environments
- Ensure consistency across systems

6. Advantages of Hyperledger Fabric
- Permissioned and secure network
- High performance and scalability
- Modular architecture
- Supports multiple programming languages

7. Application Areas
- Supply chain management
- Banking and finance
- Healthcare systems
- Digital identity management

1.6 Procedure:
The following steps were performed to design and deploy chaincode using Hyperledger Fabric:

Step 1: Start WSL (Ubuntu Environment)
- Open Command Prompt and run:

wsl
- Ubuntu terminal is launched for executing Linux commands.

Step 2: Update System Packages
sudo apt update
sudo apt upgrade

Step 3: Install Required Tools
sudo apt install git curl docker.io docker-compose golang-go jq -y

Step 4: Start Docker Service
sudo service docker start
- Verify Docker installation:

docker --version

Step 5: Configure Docker Permissions

```
sudo usermod -a -G docker $USER
newgrp docker
```

Step 6: Create Directory for Fabric Setup

```
mkdir -p $HOME/go/src/github.com
cd $HOME/go/src/github.com
```

Step 7: Download Hyperledger Fabric Installation Script

```
curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
chmod +x install-fabric.sh
```

Step 8: Install Fabric Samples, Docker Images and Binaries

```
export DOCKER_CONFIG=/tmp/docker-config
./install-fabric.sh docker samples binary
```

Step 9: Navigate to Test Network Directory

```
cd fabric-samples/test-network
```

Step 10: Start the Hyperledger Fabric Network

```
./network.sh up
```

- This starts peer nodes and orderer using Docker containers.

Step 11: Create Channel

```
./network.sh createChannel
```

- A communication channel named mychannel is created between organizations.

Step 12: Deploy Chaincode

```
./network.sh deployCC -ccn basic -ccp ../asset-transfer-basic/chaincode-go -ccl go
```

- Chaincode (smart contract) is deployed on the channel.

Step 13: Set Environment Variables

```
export PATH=${PWD}/../bin:$PATH
export FABRIC_CFG_PATH=$PWD/../config/

export CORE_PEER_TLS_ENABLED=true
export CORE_PEER_LOCALMSPID="Org1MSP"
export
CORE_PEER_TLS_ROOTCERT_FILE=${PWD}/organizations/peerOrganizations/org1.example
.com/peers/peer0.org1.example.com/tls/ca.crt
export
CORE_PEER_MSPCONFIGPATH=${PWD}/organizations/peerOrganizations/org1.example.co
m/users/Admin@org1.example.com/msp
export CORE_PEER_ADDRESS=localhost:7051
```

Step 14: Query the Ledger Data

peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'

- This command retrieves all assets stored in the blockchain ledger.

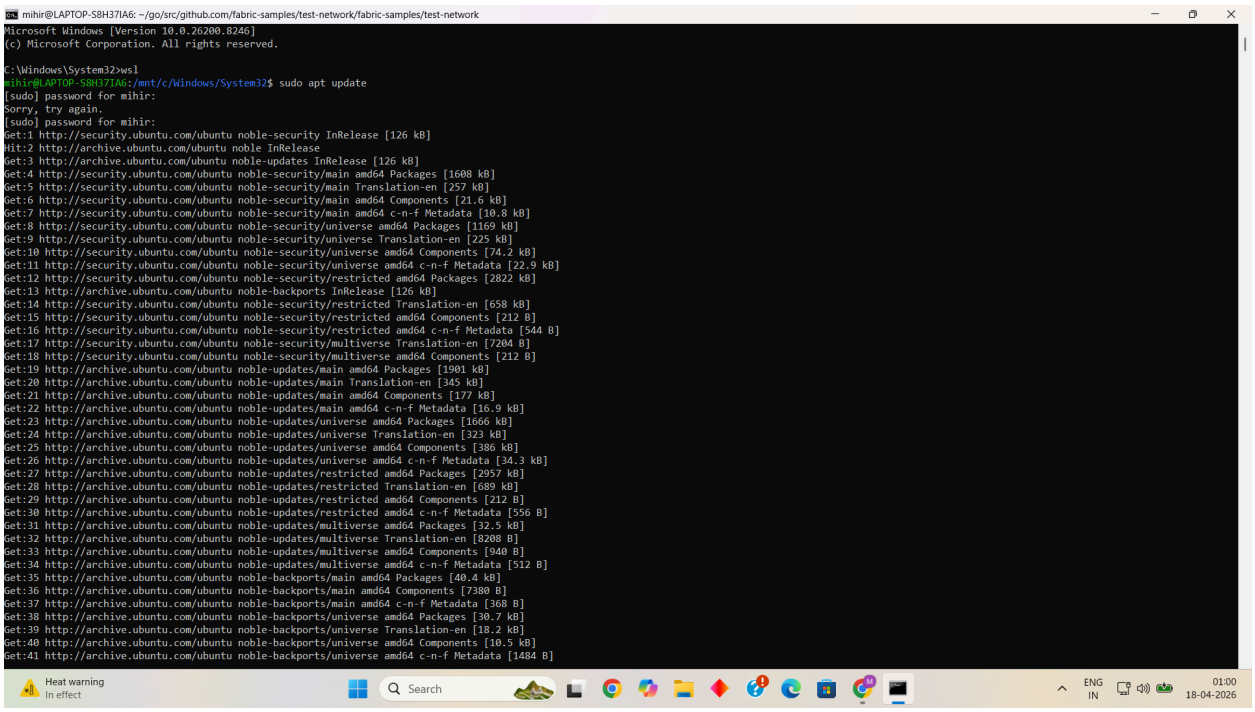
Step 15: Observe Output

- The output displays a list of assets with details such as ID, color, owner, size, and value.

Summary of Procedure

- The Fabric network was successfully started
- Channel was created
- Chaincode was deployed
- Ledger data was queried

1.7 Program and Output:



```
mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>wsl
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ sudo apt update
[sudo] password for mihir:
Sorry, try again.
[sudo] password for mihir:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1608 kB]
Get:5 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [257 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.8 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1169 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.9 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2822 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [658 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [544 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [7204 B]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:19 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1901 kB]
Get:20 http://archive.ubuntu.com/ubuntu noble-updates/main Translation-en [345 kB]
Get:21 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:22 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:23 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1666 kB]
Get:24 http://archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:25 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:26 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.3 kB]
Get:27 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2957 kB]
Get:28 http://archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [689 kB]
Get:29 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:30 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 c-n-f Metadata [556 B]
Get:31 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [32.5 kB]
Get:32 http://archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [8208 B]
Get:33 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:34 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [512 B]
Get:35 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.4 kB]
Get:36 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7380 B]
Get:37 http://archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:38 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:39 http://archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
Get:40 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:41 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1484 B]
```

mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
Get:45 http://archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 15.9 MB in 5s (3121 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
278 packages can be upgraded. Run 'apt list --upgradable' to see them.
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ sudo apt upgrade
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0
  libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  ethtool libllvm20 mesa-libgallium
The following packages have been kept back:
  libgl1-amber-dri libglapi-mesa
The following packages will be upgraded:
  apache2 apache2-bin apache2-data apache2-utils apparmor apport apport-core-dump-handler apt
  apt-utils base-files binutils binutils-common binutils-x86-64-linux-gnu bsdextrautils bsdutils
  cloud-init coreutils cpp-13 cpp-13-x86-64-linux-gnu curl dconf-gsettings-backend dconf-service
  dhcpcd-base dirmngr distro-info-data dmsetup dpkg dpkg-dev eject fdisk g++-13
  g++-13-x86-64-linux-gnu gcc-13 gcc-13-base gcc-13-x86-64-linux-gnu gcc-14-base gir1.2-glib-2.0
  gir1.2-packagekitglib-1.0 git git-man gnupg gnupg-l10n gnupg-utils gpg gpg-agent gpg-wks-client
  gpgconf gpgsm gpgv gtk-update-icon-cache gzip iproute2 iputils-ping keyboxd kmod krb5-locales
  landscape-client landscape-common libacl1 libapache2-mod-php8.3 libapparmor1 libapt-pkg6.0t64
  libasan8 libatomic1 libattr1 libaudit-common libaudit1 libavahi-client3 libavahi-common-data
  libavahi-common3 libbinutils libblkid1 libbsd0 libc-bin libc-dev-bin libc-devtools libc6 libc6-dev
  libcap2 libcap2-bin libcc1-0 libcryptsetup12 libctf-nobfd0 libctf0 libcups2t64 libcurl3t64-gnutls
  libcurl4t64 libdconf1 libdevmapper1.02.1 libdpkg-perl libdrm-amdgpu1 libdrm-common libdrm-intel1
  libdrm-nouveau2 libdrm-radeon1 libdrm2 libdw1t64 libegl-mesa0 libelf1t64 libexpat1 libexpat1-dev
  libfdisk1 libfreetype-dev libfreetype6 libgbm1 libgcc-13-dev libgcc-s1 libgdk-pixbuf-2.0-0
  libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgl1-mesa-dri libglib2.0-0t64 libglib2.0-bin
  libglib2.0-data libglx-mesa0 libgmp10 libgnutls30t64 libgomp1 libgpg-error-l10n libgpg-error0
  libgprofng0 libgssapi-krb5-2 libgstreamer1.0-0 libgtk-3-0t64 libgtk-3-bin libgtk-3-common
  libheif-dev libheif-plugin-aomdec libheif-plugin-aomenc libheif-plugin-libde265 libheif1 libhwasan0
  libidn2-0 libisl23 libitm1 libk5crypto3 libkmod2 libkrb5-3 libkrb5support0 libldap-common libldap2
  liblsan0 liblzma-dev liblzma5 libmd0 libmount1 libmpc3 libmpfr6 libnetplan1 libnghttp2-14
  libnss-systemd libnuma1 libpackagekit-glib2-18 libpam-cap libpam-modules libpam-modules-bin
  libpam-runtime libpam-systemd libpam0g libpciaccess0 libpcre2-8-0 libperl5.38t64 libpng-dev
  libpng-tools libpng16-16t64 libpolkit-agent-1-0 libpolkit-gobject-1-0 libproc2-0 libpython3-stdlib
  libpython3.12-minimal libpython3.12-stdlib libpython3.12t64 libquadmath0 libselinux1 libsframe1
  libsmartcols1 libsodium23 libsqlite3-0 libssh-4 libssl-dev libssl3t64 libstdc++-13-dev libstdc++6
  libsystemd-shared libsystemd0 libtasn1-6 libtiff-dev libtiff6 libtiffxx6 libtsan2 libubsan1 libudev1
  libunistring5 libunwind8 libuuid1 libvpx-dev libvpx9 libxml2 linux-libc-dev locales login lshw
  mesa-vulkan-drivers motd-news-config mount nano netplan-generator netplan.io openssh-client openssl
  packagekit packagekit-tools passwd pci.ids perl perl-base perl-modules-5.38 php8.3 php8.3-cli
```

Heat warning
In effect

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01:00
18-04-2026

mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
Setting up libheif-plugin-aomenc:amd64 (1.17.6-1ubuntu4.2) ...
Setting up update-manager-core (1:24.04.12) ...
Processing triggers for sgml-base (1.31) ...
Processing triggers for install-info (7.1-3build2) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Processing triggers for systemd (255.4-1ubuntu8.15) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Setting up packagekit (1.2.8-2ubuntu1.4) ...
Setting up packagekit-tools (1.2.8-2ubuntu1.4) ...
Setting up software-properties-common (0.99.49.4) ...
Setting up ubuntu-wsl (1.539.2) ...
Processing triggers for php8.3-cli (8.3.6-0ubuntu0.24.04.8) ...
Processing triggers for libapache2-mod-php8.3 (8.3.6-0ubuntu0.24.04.8) ...
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ sudo apt install git curl docker.io docker-compose -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.8).
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0
  libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base iptables libip4tc2 libip6tc2
  libnetfilter-conntrack3 libnfnetlink0 libnftables1 libnftnl11 nftables pigz python3-compose
  python3-docker python3-dockerpty python3-docopt python3-dotenv python3-packaging python3-texttable
  python3-websocket runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools btrfs-progs cgroupfs-mount | cgroup-lite debootstrap docker-buildx
  docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils firewalld
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker-compose docker.io iptables libip4tc2
  libip6tc2 libnetfilter-conntrack3 libnfnetlink0 libnftables1 libnftnl11 nftables pigz
  python3-compose python3-docker python3-dockerpty python3-docopt python3-dotenv python3-packaging
  python3-texttable python3-websocket runc ubuntu-fan
0 upgraded, 25 newly installed, 0 to remove and 2 not upgraded.
Need to get 75.1 MB of archives.
After this operation, 285 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 libip4tc2 amd64 1.8.10-3ubuntu2 [23.3 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/main amd64 libip6tc2 amd64 1.8.10-3ubuntu2 [23.7 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble/main amd64 libnfnetlink0 amd64 1.0.2-2build1 [14.8 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble/main amd64 libnetfilter-conntrack3 amd64 1.0.9-6build1 [45.2 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/main amd64 libnftnl11 amd64 1.2.6-2build1 [66.0 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/main amd64 iptables amd64 1.8.10-3ubuntu2 [381 kB]
```

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01:01
18-04-2026

mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
Created symlink /etc/systemd/system/multi-user.target.wants/containerd.service → /usr/lib/systemd/system/containerd.service.
Setting up python3-websocket (1.7.0-1) ...
Setting up python3-dockerpty (0.4.1-5) ...
Setting up libnftables1:amd64 (1.0.9-1ubuntu0.1) ...
Setting up nftables (1.0.9-1ubuntu0.1) ...
Setting up python3-docker (5.0.3-1ubuntu1.1) ...
Setting up python3-compose (1.29.2-6ubuntu1) ...
Setting up libnetfilter-conntrack3:amd64 (1.0.9-6build1) ...
Setting up docker-compose (1.29.2-6ubuntu1) ...
Setting up iptables (1.8.10-3ubuntu2) ...
update-alternatives: using /usr/sbin/iptables-legacy to provide /usr/sbin/iptables (iptables) in auto mode
update-alternatives: using /usr/sbin/ip6tables-legacy to provide /usr/sbin/ip6tables (ip6tables) in auto mode
update-alternatives: using /usr/sbin/iptables-nft to provide /usr/sbin/iptables (iptables) in auto mode
update-alternatives: using /usr/sbin/ip6tables-nft to provide /usr/sbin/ip6tables (ip6tables) in auto mode
update-alternatives: using /usr/sbin/arptables-nft to provide /usr/sbin/arptables (arptables) in auto mode
update-alternatives: using /usr/sbin/ebtables-nft to provide /usr/sbin/ebtables (ebtables) in auto mode
Setting up docker.io (29.1.3-0ubuntu3~24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Setting up dnsmasq-base (2.90-2ubuntu0.1) ...
Setting up ubuntu-fan (0.12.16+24.04.1) ...
Created symlink /etc/systemd/system/multi-user.target.wants/ubuntu-fan.service → /usr/lib/systemd/system/ubuntu-fan.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ sudo service docker start
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~24.04.1
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ sudo usermod -a -G docker $USER
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ newgrp docker
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ sudo apt install golang-go -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0
  libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  golang-1.22-go golang-1.22-src golang-src
Suggested packages:
  bzr | brz mercurial subversion
The following NEW packages will be installed:
  golang-1.22-go golang-1.22-src golang-go golang-src
0 upgraded, 4 newly installed, 0 to remove and 2 not upgraded.
Need to get 45.7 MB of archives.
After this operation, 228 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 golang-1.22-src all 1.22.2-2ubuntu0.4 [19.7 MB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 golang-1.22-go amd64 1.22.2-2ubuntu0.4 [25.9 MB]
```

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01:01
18-04-2026

mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
Fetched 45.7 MB in 5s (8540 kB/s)
Selecting previously unselected package golang-1.22-src.
(Reading database ... 49718 files and directories currently installed.)
Preparing to unpack .../golang-1.22-src_1.22.2-2ubuntu0.4_all.deb ...
Unpacking golang-1.22-src (1.22.2-2ubuntu0.4) ...
Selecting previously unselected package golang-1.22-go.
Preparing to unpack .../golang-1.22-go_1.22.2-2ubuntu0.4_amd64.deb ...
Unpacking golang-1.22-go (1.22.2-2ubuntu0.4) ...
Selecting previously unselected package golang-src.
Preparing to unpack .../golang-src_2%3a1.22~2build1_all.deb ...
Unpacking golang-src (2:1.22~2build1) ...
Selecting previously unselected package golang-go:amd64.
Preparing to unpack .../golang-go_2%3a1.22~2build1_amd64.deb ...
Unpacking golang-go:amd64 (2:1.22~2build1) ...
Setting up golang-1.22-src (1.22.2-2ubuntu0.4) ...
Setting up golang-src (2:1.22~2build1) ...
Setting up golang-1.22-go (1.22.2-2ubuntu0.4) ...
Setting up golang-go:amd64 (2:1.22~2build1) ...
Processing triggers for man-db (2.12.0-4build2) ...
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ go version
go version go1.22.2 linux/amd64
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ mkdir -p $HOME/go/src/github.com
mihir@LAPTOP-S8H37IA6:/mnt/c/Windows/System32$ cd $HOME/go/src/github.com
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ chmod +x install-fabric.sh
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ ./install-fabric.sh docker samples

Clone hyperledger/fabric-samples repo

===> Cloning hyperledger/fabric-samples repo
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15633, done.
remote: Counting objects: 100% (239/239), done.
remote: Compressing objects: 100% (123/123), done.
remote: Total 15633 (delta 175), reused 116 (delta 116), pack-reused 15394 (from 3)
Receiving objects: 100% (15633/15633), 24.07 MiB | 9.61 MiB/s, done.
Resolving deltas: 100% (8824/8824), done.
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccenv baseos
===> Pulling fabric Images
====>  ghcr.io/hyperledger/fabric-peer:2.5.15
error getting credentials - err: fork/exec /mnt/c/Program Files/Docker/Docker/resources/bin/docker-credential-desktop.exe: exec format error, out: ``
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-peer:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-peer:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-peer:2.5.15
====>  ghcr.io/hyperledger/fabric-orderer:2.5.15
```

Heat warning
In effect

Search

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IN

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mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-peer:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-peer:2.5.15
====>  ghcr.io/hyperledger/fabric-orderer:2.5.15
error getting credentials - err: fork/exec /mnt/c/Program Files/Docker/Docker/resources/bin/docker-credential-desktop.exe: exec format error, out: ``
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-orderer:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-orderer:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-orderer:2.5.15
====>  ghcr.io/hyperledger/fabric-ccenv:2.5.15
error getting credentials - err: fork/exec /mnt/c/Program Files/Docker/Docker/resources/bin/docker-credential-desktop.exe: exec format error, out: ``
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-ccenv:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-ccenv:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-ccenv:2.5.15
====>  ghcr.io/hyperledger/fabric-baseos:2.5.15
error getting credentials - err: fork/exec /mnt/c/Program Files/Docker/Docker/resources/bin/docker-credential-desktop.exe: exec format error, out: ``
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-baseos:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-baseos:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-baseos:2.5.15
===> Pulling fabric ca Image
====>  ghcr.io/hyperledger/fabric-ca:1.5.17
error getting credentials - err: fork/exec /mnt/c/Program Files/Docker/Docker/resources/bin/docker-credential-desktop.exe: exec format error, out: ``
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-ca:1.5.17
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-ca:1.5.17
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-ca:1.5.17
===> List out hyperledger images
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
chmod +x install-fabric.sh
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ ./install-fabric.sh docker samples

Clone hyperledger/fabric-samples repo

===> Changing directory to fabric-samples
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccenv baseos
===> Pulling fabric Images
====>  ghcr.io/hyperledger/fabric-peer:2.5.15
error getting credentials - err: fork/exec /mnt/c/Program Files/Docker/Docker/resources/bin/docker-credential-desktop.exe: exec format error, out: ``
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-peer:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-peer:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-peer:2.5.15
====>  ghcr.io/hyperledger/fabric-orderer:2.5.15
error getting credentials - err: fork/exec /mnt/c/Program Files/Docker/Docker/resources/bin/docker-credential-desktop.exe: exec format error, out: ``
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-orderer:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-orderer:2.5.15
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-orderer:2.5.15
====>  ghcr.io/hyperledger/fabric-ccenv:2.5.15
```

Heat warning
In effect
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mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
error getting credentials - err: fork/exec /mnt/c/Program Files/Docker/Docker/resources/bin/docker-credential-desktop.exe: exec format error, out: ``
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-ca:1.5.17
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-ca:1.5.17
Error response from daemon: No such image: ghcr.io/hyperledger/fabric-ca:1.5.17
===> List out hyperledger images
WARNING: This output is designed for human readability. For machine-readable output, please use --format.
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ rm ~/.docker/config.json
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ sudo service docker restart
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ cd ~/go/src/github.com
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ ./install-fabric.sh docker samples

Clone hyperledger/fabric-samples repo

===> Changing directory to fabric-samples
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric docker images

FABRIC_IMAGES: peer orderer ccenv baseos
===> Pulling fabric Images
```

Heat warning
In effect
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mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
hyperledger/fabric-ccenv:2.5.15             86dded7eda92          1GB          255MB
hyperledger/fabric-ccenv:latest             86dded7eda92          1GB          255MB
hyperledger/fabric-orderer:2.5              9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:2.5.15           9972c209be47        178MB         49.5MB
hyperledger/fabric-orderer:latest           9972c209be47        178MB         49.5MB
hyperledger/fabric-peer:2.5                 3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:2.5.15              3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:latest              3e25adc31a75        230MB         66.2MB
mihir@LAPTOP-S8H37IA6:~/go/src/github.com$ cd fabric-samples/test-network
./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb' with crypto from 'cryptogen'
Peer binary and configuration files not found..

Follow the instructions in the Fabric docs to install the Fabric Binaries:
https://hyperledger-fabric.readthedocs.io/en/latest/install.html
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network$ rm -rf fabric-samples
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network$ export DOCKER_CONFIG=/tmp/docker-config
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network$ curl -sSLO https://raw.githubusercontent.com/hyperledger/fabric/main/scripts/install-fabric.sh
chmod +x install-fabric.sh
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network$ ./install-fabric.sh docker samples binary

Clone hyperledger/fabric-samples repo

===> Cloning hyperledger/fabric-samples repo
Cloning into 'fabric-samples'...
remote: Enumerating objects: 15633, done.
remote: Counting objects: 100% (239/239), done.
remote: Compressing objects: 100% (123/123), done.
remote: Total 15633 (delta 175), reused 116 (delta 116), pack-reused 15394 (from 3)
Receiving objects: 100% (15633/15633), 24.07 MiB | 17.25 MiB/s, done.
Resolving deltas: 100% (8824/8824), done.
fabric-samples v2.5.15 does not exist, defaulting to main. fabric-samples main branch is intended to work with recent versions of fabric.

Pull Hyperledger Fabric binaries

===> Downloading version 2.5.15 platform specific fabric binaries
===> Downloading:  https://github.com/hyperledger/fabric/releases/download/v2.5.15/hyperledger-fabric-linux-amd64-2.5.15.tar.gz
===> Will unpack to: /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:--  --:--:--  --:--:--     0
100  120M  100  120M    0     0  15.3M      0  0:00:07  0:00:07 --:--:-- 20.5M
==> Done.
===> Downloading version 1.5.17 platform specific fabric-ca-client binary
===> Downloading:  https://github.com/hyperledger/fabric-ca/releases/download/v1.5.17/hyperledger-fabric-ca-linux-amd64-1.5.17.tar.gz
===> Will unpack to: /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
```

Heat warning
In effect
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mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
hyperledger/fabric-peer:2.5                 3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:2.5.15              3e25adc31a75        230MB         66.2MB
hyperledger/fabric-peer:latest              3e25adc31a75        230MB         66.2MB
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network$ cd fabric-samples/test-network
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network$ ./network.sh up
Using docker and docker-compose
Starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb' with crypto from 'cryptogen'
LOCAL_VERSION=v2.5.15
DOCKER_IMAGE_VERSION=v2.5.15
/home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/../bin/cryptogen
Generating certificates using cryptogen tool
Creating Org1 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org1.yaml --output=organizations
org1.example.com
+ res=0
Creating Org2 Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-org2.yaml --output=organizations
org2.example.com
+ res=0
Creating Orderer Org Identities
+ cryptogen generate --config=./organizations/cryptogen/crypto-config-orderer.yaml --output=organizations
+ res=0
Generating CCP files for Org1 and Org2
Creating network "fabric_test" with the default driver
Creating volume "compose_orderer.example.com" with default driver
Creating volume "compose_peer0.org1.example.com" with default driver
Creating volume "compose_peer0.org2.example.com" with default driver
Creating orderer.example.com    ... done
Creating peer0.org1.example.com ... done
Creating peer0.org2.example.com ... done
CONTAINER ID   IMAGE                               COMMAND             CREATED         STATUS                     PORTS
                                                   NAMES
528ec77029bd   hyperledger/fabric-peer:latest      "peer node start"   2 seconds ago   Up Less than a second      0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
                                                   peer0.org1.example.com
a4d4ff5e38e8   hyperledger/fabric-peer:latest      "peer node start"   2 seconds ago   Up Less than a second      0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tc
p                                                  peer0.org2.example.com
b888e27700d9   hyperledger/fabric-orderer:latest   "orderer"           2 seconds ago   Up Less than a second      0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0
:9443->9443/tcp, [::]:9443->9443/tcp   orderer.example.com
5b5d006efbb7   hello-world                         "/hello"            4 minutes ago   Exited (0) 4 minutes ago
                                                   exciting_dubinsky
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network$ ./network.sh createChannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb
Bringing up network
LOCAL_VERSION=v2.5.15
DOCKER_IMAGE_VERSION=v2.5.15
peer0.org2.example.com is up-to-date
orderer.example.com is up-to-date
```

Heat warning
In effect
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mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
        "consensusRelation": "consenter",
        "status": "active",
        "height": 1
}

Channel 'mychannel' created
Joining org1 peer to the channel...
Using organization 1
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
2026-04-17 19:24:32.067 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-04-17 19:24:32.115 UTC 0002 INFO [channelCmd] executeJoin -> Successfully submitted proposal to join channel
Joining org2 peer to the channel...
Using organization 2
+ peer channel join -b ./channel-artifacts/mychannel.block
+ res=0
2026-04-17 19:24:37.270 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-04-17 19:24:37.308 UTC 0002 INFO [channelCmd] executeJoin -> Successfully submitted proposal to join channel
Setting anchor peer for org1...
Using organization 1
Fetching channel config for channel mychannel
Using organization 1
Fetching the most recent configuration block for the channel
++ peer channel fetch config /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/channel-artifacts/config_block.pb -o localhost:7050 --ordererTLSHostnameOverride orderer.example
.com -c mychannel --tls --cafile /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem
2026-04-17 19:24:37.382 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-04-17 19:24:37.385 UTC 0002 INFO [cli.common] readBlock -> Received block: 0
2026-04-17 19:24:37.385 UTC 0003 INFO [channelCmd] fetch -> Retrieving last config block: 0
2026-04-17 19:24:37.387 UTC 0004 INFO [cli.common] readBlock -> Received block: 0
Decoding config block to JSON and isolating config to /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/channel-artifacts/Org1MSPconfig.json
++ configtxlator proto_decode --input /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/channel-artifacts/config_block.pb --type common.Block --output /home/mihir/go/src/githu
b.com/fabric-samples/test-network/fabric-samples/test-network/channel-artifacts/config_block.json
++ jq '.data.data[0].payload.data.config' /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/channel-artifacts/config_block.json
/home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/scripts/configUpdate.sh: line 35: jq: command not found
++ res=127
Failed to parse channel configuration, make sure you have jq installed
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network$ sudo apt install jq -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libwayland-server0
  libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libjq1 libonig5
The following NEW packages will be installed:
  jq libjq1 libonig5
0 upgraded, 3 newly installed, 0 to remove and 2 not upgraded.
```

Heat warning
In effect

Search

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mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network$ ./network.sh createChannel
Using docker and docker-compose
Creating channel 'mychannel'.
If network is not up, starting nodes with CLI timeout of '5' tries and CLI delay of '3' seconds and using database 'leveldb'
Bringing up network
LOCAL_VERSION=v2.5.15
DOCKER_IMAGE_VERSION=v2.5.15
peer0.org1.example.com is up-to-date
orderer.example.com is up-to-date
peer0.org2.example.com is up-to-date
CONTAINER ID   IMAGE                               COMMAND             CREATED         STATUS                     PORTS
                                                   NAMES
528ec77029bd   hyperledger/fabric-peer:latest      "peer node start"   2 minutes ago   Up 2 minutes               0.0.0.0:7051->7051/tcp, [::]:7051->7051/tcp, 0.0.0.0:9444->9444/tcp, [::]:9444->9444/tcp
                                                   peer0.org1.example.com
a4d4ff5e38e8   hyperledger/fabric-peer:latest      "peer node start"   2 minutes ago   Up 2 minutes               0.0.0.0:9051->9051/tcp, [::]:9051->9051/tcp, 7051/tcp, 0.0.0.0:9445->9445/tcp, [::]:9445->9445/tc
p                                                  peer0.org2.example.com
b888e27700d9   hyperledger/fabric-orderer:latest   "orderer"           2 minutes ago   Up 2 minutes               0.0.0.0:7050->7050/tcp, [::]:7050->7050/tcp, 0.0.0.0:7053->7053/tcp, [::]:7053->7053/tcp, 0.0.0.0
:9443->9443/tcp, [::]:9443->9443/tcp   orderer.example.com
5b5d006efbb7   hello-world                         "/hello"            6 minutes ago   Exited (0) 6 minutes ago
                                                   exciting_dubinsky
Using docker and docker-compose
Generating channel genesis block 'mychannel.block'
Using organization 1
/home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/../bin/configtxgen
+ '[' 0 -eq 1 ']'
+ configtxgen -profile ChannelUsingRaft -outputBlock ./channel-artifacts/mychannel.block -channelID mychannel
2026-04-17 19:26:05.994 UTC 0001 INFO [common.tools.configtxgen] main -> Loading configuration
2026-04-17 19:26:05.997 UTC 0002 INFO [common.tools.configtxgen.localconfig] completeInitialization -> orderer type: etcdraft
2026-04-17 19:26:05.997 UTC 0003 INFO [common.tools.configtxgen.localconfig] completeInitialization -> Orderer.EtcdRaft.Options unset, setting to tick_interval:"500ms" election_tick:10 heartbeat_tick:1 max_infli
ght_blocks:5 snapshot_interval_size:16777216
2026-04-17 19:26:05.997 UTC 0004 INFO [common.tools.configtxgen.localconfig] Load -> Loaded configuration: /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/configtx/configtx.
yaml
2026-04-17 19:26:06.000 UTC 0005 INFO [common.tools.configtxgen] doOutputBlock -> Generating genesis block
2026-04-17 19:26:06.000 UTC 0006 INFO [common.tools.configtxgen] doOutputBlock -> Creating application channel genesis block
2026-04-17 19:26:06.000 UTC 0007 INFO [common.tools.configtxgen] doOutputBlock -> Writing genesis block
+ res=0
Creating channel mychannel
Adding orderers
+ . scripts/orderer.sh mychannel
+ '[' 0 -eq 1 ']'
+ res=0
2026-04-17 19:24:37.270 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-04-17 19:24:37.308 UTC 0002 INFO [channelCmd] executeJoin -> Successfully submitted proposal to join channel
Status: 405
{
        "error": "cannot join: channel already exists"
}
```

Heat warning
In effect

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mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
Error: proposal failed (err: bad proposal response 500: cannot create ledger from genesis block: ledger [mychannel] already exists with state [ACTIVE])
After 5 attempts, peer0.org1 has failed to join channel 'mychannel'
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network$ ./netmihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network$ ./networ
k.sh deployCC -ccn basic -ccp ../asset-transfer-basic/chaincode-go -ccl go
Using docker and docker-compose
deploying chaincode on channel 'mychannel'
executing with the following
- CHANNEL_NAME: mychannel
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-go
- CC_SRC_LANGUAGE: go
- CC_VERSION: 1.0
- CC_SEQUENCE: auto
- CC_END_POLICY: NA
- CC_COLL_CONFIG: NA
- CC_INIT_FCN: NA
- DELAY: 3
- MAX_RETRY: 5
- VERBOSE: false
executing with the following
- CC_NAME: basic
- CC_SRC_PATH: ../asset-transfer-basic/chaincode-go
- CC_SRC_LANGUAGE: go
- CC_VERSION: 1.0
Vendoring Go dependencies at ../asset-transfer-basic/chaincode-go
~/go/src/github.com/fabric-samples/test-network/fabric-samples/asset-transfer-basic/chaincode-go ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network
go: downloading go1.23.0 (linux/amd64)
go: downloading github.com/hyperledger/fabric-protos-go-apiv2 v0.3.4
go: downloading github.com/hyperledger/fabric-contract-api-go/v2 v2.2.0
go: downloading github.com/hyperledger/fabric-chaincode-go/v2 v2.0.0
go: downloading github.com/stretchr/testify v1.10.0
go: downloading google.golang.org/protobuf v1.36.4
go: downloading github.com/davecgh/go-spew v1.1.1
go: downloading github.com/pmezard/go-difflib v1.0.0
go: downloading gopkg.in/yaml.v3 v3.0.1
go: downloading google.golang.org/grpc v1.71.0
go: downloading github.com/xeipuuv/gojsonschema v1.2.0
go: downloading github.com/go-openapi/spec v0.21.0
go: downloading golang.org/x/net v0.38.0
go: downloading github.com/xeipuuv/gojsonreference v0.0.0-20180127040603-bd5ef7bd5415
go: downloading google.golang.org/genproto v0.0.0-20230410155749-daa745c078e1
go: downloading golang.org/x/sys v0.31.0
go: downloading github.com/xeipuuv/gojsonpointer v0.0.0-20190905194746-02993c407bfb
go: downloading github.com/go-openapi/swag v0.23.0
go: downloading github.com/go-openapi/jsonpointer v0.21.0
go: downloading github.com/go-openapi/jsonreference v0.21.0
go: downloading github.com/mailru/easyjson v0.7.7
go: downloading golang.org/x/text v0.23.0
go: downloading github.com/josharian/intern v1.0.0
```

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mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network

```
Checking the commit readiness of the chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Checking the commit readiness of the chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to check the commit readiness of the chaincode definition on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode checkcommitreadiness --channelID mychannel --name basic --version 1.0 --sequence 1 --output json
+ res=0
{
        "approvals": {
                "Org1MSP": true,
                "Org2MSP": false
        }
}
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'
Using organization 2
+ peer lifecycle chaincode approveformyorg -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/
organizations/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --version 1.0 --package-id basic_1.0:1f66cc610fb8ac88d69e78edc4c1cb09bac7c839fb5399009310ac8dfa7
03799 --sequence 1
+ res=0
2026-04-17 19:29:46.569 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [497e867f453c434e5a7e3fe1b6460842a81dc902f0f6b1361e6b0e4f3ee9efa9] committed with status (VALID) at localhost:9051
Chaincode definition approved on peer0.org2 on channel 'mychannel'
Using organization 1
Checking the commit readiness of the chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to check the commit readiness of the chaincode definition on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode checkcommitreadiness --channelID mychannel --name basic --version 1.0 --sequence 1 --output json
+ res=0
{
        "approvals": {
                "Org1MSP": true,
                "Org2MSP": true
        }
}
Checking the commit readiness of the chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Checking the commit readiness of the chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to check the commit readiness of the chaincode definition on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode checkcommitreadiness --channelID mychannel --name basic --version 1.0 --sequence 1 --output json
+ res=0
{
        "approvals": {
                "Org1MSP": true,
                "Org2MSP": true
        }
}
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'
Using organization 1
Using organization 2
+ peer lifecycle chaincode commit -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/organizat
ions/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --peerAddresses localhost:7051 --tlsRootCertFiles /home/mihir/go/src/github.com/fabric-samples/test-netwo
rk/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /home/mihir/go/src/github.com/fabric-sample
```

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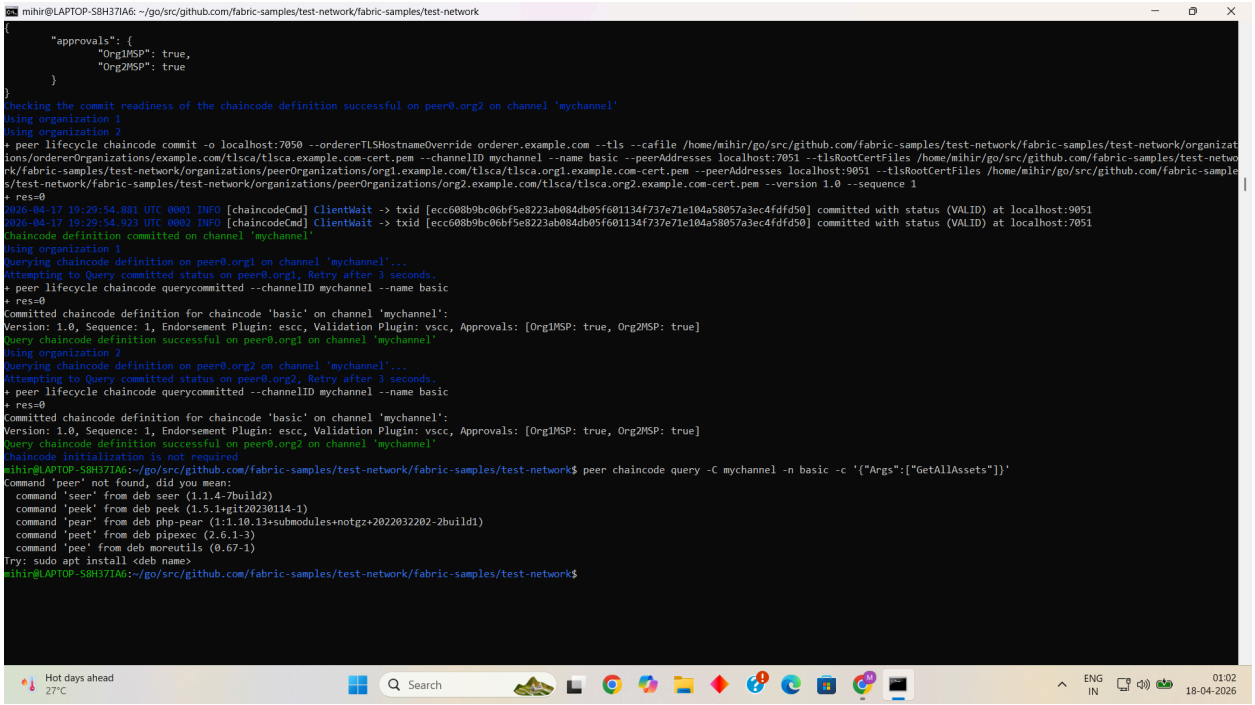

```
mihir@LAPTOP-S8H37IA6: ~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network
{
        "approvals": {
                "Org1MSP": true,
                "Org2MSP": true
        }
}
Checking the commit readiness of the chaincode definition successful on peer0.org2 on channel 'mychannel'
Using organization 1
Using organization 2
+ peer lifecycle chaincode commit -o localhost:7050 --ordererTLSHostnameOverride orderer.example.com --tls --cafile /home/mihir/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network/organizat
ions/ordererOrganizations/example.com/tlsca/tlsca.example.com-cert.pem --channelID mychannel --name basic --peerAddresses localhost:7051 --tlsRootCertFiles /home/mihir/go/src/github.com/fabric-samples/test-netwo
rk/fabric-samples/test-network/organizations/peerOrganizations/org1.example.com/tlsca/tlsca.org1.example.com-cert.pem --peerAddresses localhost:9051 --tlsRootCertFiles /home/mihir/go/src/github.com/fabric-sample
s/test-network/fabric-samples/test-network/organizations/peerOrganizations/org2.example.com/tlsca/tlsca.org2.example.com-cert.pem --version 1.0 --sequence 1
+ res=0
2026-04-17 19:29:54.881 UTC 0001 INFO [chaincodeCmd] ClientWait -> txid [ecc608b9bc06bf5e8223ab084db05f601134f737e71e104a58057a3ec4fdfd50] committed with status (VALID) at localhost:9051
2026-04-17 19:29:54.923 UTC 0002 INFO [chaincodeCmd] ClientWait -> txid [ecc608b9bc06bf5e8223ab084db05f601134f737e71e104a58057a3ec4fdfd50] committed with status (VALID) at localhost:7051
Chaincode definition committed on channel 'mychannel'
Using organization 1
Querying chaincode definition on peer0.org1 on channel 'mychannel'...
Attempting to Query committed status on peer0.org1, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org1 on channel 'mychannel'
Using organization 2
Querying chaincode definition on peer0.org2 on channel 'mychannel'...
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic
+ res=0
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vscc, Approvals: [Org1MSP: true, Org2MSP: true]
Query chaincode definition successful on peer0.org2 on channel 'mychannel'
Chaincode initialization is not required
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network$ peer chaincode query -C mychannel -n basic -c '{"Args":["GetAllAssets"]}'
Command 'peer' not found, did you mean:
  command 'seer' from deb seer (1.1.4-7build2)
  command 'peek' from deb peek (1.5.1+git20230114-1)
  command 'pear' from deb php-pear (1:1.10.13+submodules+notgz+2022032202-2build1)
  command 'peet' from deb pipexec (2.6.1-3)
  command 'pee' from deb moreutils (0.67-1)
Try: sudo apt install <deb name>
mihir@LAPTOP-S8H37IA6:~/go/src/github.com/fabric-samples/test-network/fabric-samples/test-network$
```

1.8 Conclusion: The experiment successfully demonstrated the setup of a Hyperledger Fabric network and the deployment of chaincode in a permissioned blockchain environment. A channel was created, and smart contract functionality was implemented using chaincode to manage assets. The successful querying of ledger data confirms that transactions were executed and stored correctly. Thus, the objective of designing and deploying chaincode on a Hyperledger Fabric network was achieved.